TYPES OF KEYS

Database Management System (DBMS)

Primary • Super • Candidate • Alternate • Unique • Composite • Foreign

Concepts, examples, and SQL syntax

LECTURE SLIDES

What is a Key?

A key is a column or set of columns used to identify records or create relationships between tables.

Keys help maintain data accuracy, avoid unwanted duplicates, and connect related information.

Example table: STUDENT(StudentID, Email, Name, DeptID)

Different key types describe different rules for uniqueness and relationships.

```
STUDENT(StudentID, Email, Name, DeptID)     |     DEPARTMENT(DeptID, DeptName)
```

1. Primary Key

Uniquely identifies each row in a table.

Cannot contain NULL values, and duplicate values are not allowed.

A table has only one primary-key constraint, though it may contain multiple columns.

StudentID	Name	Email
101	Amina	amina@example.com
102	Rahim	rahim@example.com
103	Nila	nila@example.com

StudentID is the Primary Key

```
CREATE TABLE Student (StudentID INT PRIMARY KEY, Name VARCHAR(50));
```

2. Super Key and 3. Candidate Key

SUPER KEY

Any column or combination of columns that uniquely identifies a row. It may include extra, unnecessary columns.

Examples: {StudentID}, {Email}, {StudentID, Name}

CANDIDATE KEY

A minimal super key: it uniquely identifies a row and has no unnecessary columns.

Examples: {StudentID}, {Email} (if both are unique)

```
If StudentID and Email are each unique: {StudentID, Name} is a super key, but not a candidate key.
```

4. Alternate Key and 5. Unique Key

ALTERNATE KEY

A candidate key that was not selected as the primary key.

Example: StudentID is the primary key; Email is an alternate key.

UNIQUE KEY

A database constraint that prevents duplicate values in a column or column combination.

Note: NULL handling for UNIQUE constraints depends on the DBMS.

```
CREATE TABLE Student (StudentID INT PRIMARY KEY, Email VARCHAR(100) UNIQUE);
```

6. Composite Key

A key made from two or more columns together.

Used when one column alone cannot uniquely identify a row.

Example: a student can enroll in many courses, and each course has many students.

StudentID	CourseID	Semester
101	CS101	Fall
101	DB202	Fall
102	CS101	Fall

StudentID + CourseID identify each enrollment.

```
PRIMARY KEY (StudentID, CourseID)
```

7. Foreign Key

A column (or set of columns) that references a primary key or an eligible unique key in another table.

Maintains referential integrity: a child record cannot reference a parent value that does not exist (subject to the constraint rules).

Example: Student.DeptID references Department.DeptID.

DEPARTMENT: DeptID	DeptName
10	CSE
20	EEE

STUDENT: StudentID	Name	DeptID
101	Amina	10
102	Rahim	20

```
FOREIGN KEY (DeptID) REFERENCES Department(DeptID)
```

Putting Keys Together: SQL Example

A simple design for departments and students

```
CREATE TABLE Department (
  DeptID INT PRIMARY KEY,
DeptName VARCHAR(50) UNIQUE
            );
  CREATE TABLE Student (
 StudentID INT PRIMARY KEY,
 Email VARCHAR(100) UNIQUE,
 Name VARCHAR(50) NOT NULL,
        DeptID INT,
FOREIGN KEY (DeptID) REFERENCES Department(DeptID)
            );
```

Here: DeptID = primary key in Department and foreign key in Student; Email = unique key.

Quick Comparison

Key Type	Main Idea	Example
Primary	Chosen key; unique and NOT NULL	StudentID
Super	Any unique column set; may have extras	StudentID + Name
Candidate	Minimal super key	StudentID or Email
Alternate	Candidate key not chosen as primary	Email
Unique	Constraint blocks duplicates	Email UNIQUE
Composite	Two or more columns form a key	StudentID + CourseID
Foreign	References a key in another table	Student.DeptID

Practice Questions & Review

1. In a Student table, StudentID and Email are both unique. If StudentID is selected as the primary key, what is Email?

2. Why is {StudentID, Name} a super key but not a candidate key when StudentID alone is unique?

3. In an Enrollment table, why might (StudentID, CourseID) be a composite key?

4. Which key connects Student to Department?

Answers: 1) Alternate key 2) Name is unnecessary 3) Together they identify each enrollment 4) Foreign key